

Chenyu Fang, Ph.D.

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SUMMARY

Physics Ph.D. with 3+ years of applied ML/research experience in rare-event classification, time-series forecasting with calibrated uncertainty, and LLM evaluation. Develops experiments, benchmarks and reproducible ML workflows; hands-on with model distillation, LoRA fine-tuning and deployed AI applications.

TECHNICAL SKILLS

ML: Python, C++, SQL, PyTorch, scikit-learn, XGBoost, LightGBM, graph neural networks, transformers

Methods: Experimental design, bootstrap inference, importance weighting, uncertainty quantification, LLM evaluation

AI and deployment: LoRA/PEFT, model distillation, RAG, FastAPI, Docker, ONNX, AWS/GCP/Azure, CI/CD

RESEARCH EXPERIENCE

University of Oklahoma

Norman, OK

Research Assistant (part-time)

02/2026 - Present

Rare-Signal Benchmark: XGBoost vs. LorentzNet vs. Particle Transformer

Code: github.com/chenyufang-data/bsm-higgs-ml-benchmark

- Extended my published analysis with new background simulations; replaced random accept/reject tagging with probability weights validated against direct tagging, keeping yields unbiased with a 9-19x larger effective sample size.
- Compared XGBoost with a re-implemented LorentzNet, a Particle Transformer (ParT) and a Deep Sets control under pre-registered rules, three seeds and paired bootstraps; ML beat the cut-based selection by **9-21% in significance**.
- LorentzNet matched ParT at a tenth of the size while Deep Sets fell below XGBoost, evidence that relational structure drives the gain; at the higher mass, limited simulation statistics, not the model, cap the result.

PhD Researcher

08/2018 - 12/2025

Higgs Boson Rare-Signal Detection

Code: github.com/chenyufang-data/hep-ml-pipeline

- Developed Python/C++ pipelines for multi-million-event simulated datasets; engineered domain-informed features and trained XGBoost classifiers to separate rare signal from backgrounds about $1e7$ times larger.
- Evaluated classifiers with sample-weighted ROC/AUC and discovery significance, choosing thresholds on stable validation plateaus; the final model recovers **41.2% of signal at a 7.7% false-positive rate, a projected 6.2σ result**.
- Interpreted the model with SHAP and ablations; packaged it as a CLI with 69 tests and versioned releases.

Research outputs: Two first-author particle-physics preprints (2025): [arXiv:2511.08526](https://arxiv.org/abs/2511.08526) and [arXiv:2511.19604](https://arxiv.org/abs/2511.19604).

SELECTED PROJECTS

AI Tender Evaluation Assistant

02/2026 - Present

Team project, Techlent Machine Learning Engineer Fellowship (training program)

Demo: tender-eval-ai.github.io/tender-evaluation-assistant | Code: github.com/tender-eval-ai/tender-evaluation-assistant

- Co-built an LLM bid-review system with a second engineer and owned its backend and bid checks: models only read, a rule engine decides, and every value is cited to its page and checked verbatim or by an independent second read.
- Chose the simpler of two job runners after both passed a 12-scenario fault-injection harness; on a synthetic tender, rule drafting matched 15/15 checklist items.

Gridcast | NYISO Day-Ahead Load Forecasting

Demo: gridcast.cyfang.org | Code: github.com/chenyufang-data/gridcast

- Trained a global Temporal Fusion Transformer for 11 NYISO zones plus the statewide total; served as ONNX on GCP.
- Cut average hourly error (MAPE) 53% below the best naive baseline in a 12-month backtest, on par with NYISO's own day-ahead forecast (**4.80% vs. 4.85%**; difference not significant under a paired bootstrap).
- Conformal calibration raised how often actual load fell inside LightGBM's 80% forecast range from 57% to 77%.

MLE/AIE Interview Coach

Demo: coach.cyfang.org | Code: github.com/chenyufang-data/mle-aie-interview-coach

- Evaluated LoRA-tuned Qwen3-4B on 598 Claude-labeled synthetic answers with question-grouped splits (~20% held out per seed); agreement with Claude reached **0.94 on a 0-1 scale (QWK)** on five seeds vs. 0.79 for the best baseline.

EDUCATION

University of Oklahoma - PhD in Physics

Norman, OK | 08/2018 - 12/2025

Huazhong University of Science and Technology - B.S. in Physics

Wuhan, China | 09/2014 - 06/2018

Honors: Avenir Graduate Fellowship (2023, 2024). **Talks:** Parallel talk, PHENO 2025 (particle-physics conference)